EWB AWS VPC Peering Assessment

Inter-VPC connectivity using VPC peering, route tables and bastion-based SSH testing

Name	kesava navya n	AWS Region	ap-southeast-2 (Sydney)
Batch	1	Completion date	30 October 2026
Project	EWB AWS VPC Peering Assessment	AWS account	097346687370

1. Objective and Architecture

Thisreport documentsa VPC peeringexercise in which two isolated VPCs with non-overlapping CIDR blocks are connected so that private instances can communicate over private IP addresses. Evidence consists of AWS console screenshots and terminal sessions supplied for the assessment; every identifier, address.

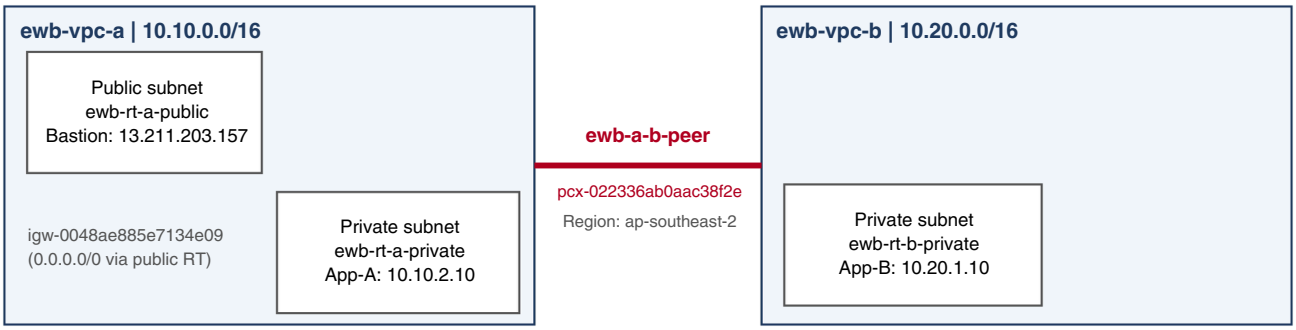


Figure A. Logical architecture reconstructed from the evidence (subnet placement of instances is inferred from route-table names and terminal output; subnet CIDR blocks are not shown in the supplied screenshots).

2. CIDR Allocation and Resource Inventory

Resource	Name / ID	CIDR / Address	Source Fig.
VPC A	ewb-vpc-a / vpc-0f69c7b9cddb668d3	10.10.0.0/16	3
VPC B	ewb-vpc-b / vpc-0fe5922e4d34e603c	10.20.0.0/16	Fig. 3
Peering connection	ewb-a-b-peer / pcx-022336ab0aac38f2e	A (requester) to B (accepter)	Fig. 3
Internet gateway	igw-0048ae885e7134e09	0.0.0.0/0 in public RT	Fig. 5
Bastion host	ewb-bastion / i-03d3560204c4a9d95	Public IPv4 13.211.203.157	Fig. 2
App-A	ewb-app-a / i-0976ea268044c3c0b ewb-	10.10.2.10 (terminal prompt)	Figs. 2, 7, 9
App-B	app-b / i-019639e2f26699b9a	10.20.1.10 (terminal prompt)	Figs. 2, 8

The two /16 ranges do not overlap, which is a precondition for VPC peering. Both VPCs, the peering connection and the instances are in **ap-southeast-2**, and requester and accepter belong to the same account (097346687370), so the peering is an intra-region, same-account connection.

3. Peering, Routing, Security Groups and Test Method

VPC peering creates a private,non-transitive link between two VPCs; it only becomes usable once both sides route to it. **Routing** was configured per route table: the A-side tables send 10.20.0.0/16 to the peering connection, the B-side table sends 10.10.0.0/16 to it, and each retains its ***local*** route. The public route table additionally holds a default route to the internet gateway, so only the bastion is reachable from outside. **Security groups** must permit ICMP between the two CIDRs and SSH (TCP 22) from the bastion; because no security-group screenshot was supplied, no group IDs or rules are asserted here **[MISSING EVIDENCE: security-group configuration]**. **Connectivity** was verified from a Windows client using an SSH ProxyJump (bastion as jump host) to reach the private instances, then ICMP pings in both directions, followed by a controlled failure test.

4. Configuration Evidence (Checkpoints 01–04)

Checkpoint 01 – Route Tables with Explicit Subnet Associations

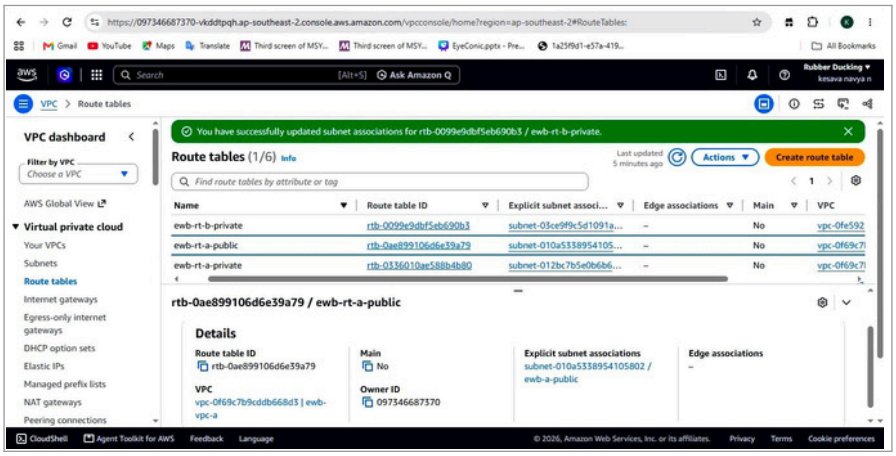


Figure 1. Checkpoint 01 – VPC console, Route tables (ap-southeast-2). Banner confirms updated subnet associations for rtb-0099e9dbf5eb690b3 / ewb-rt-b-private; ewb-rt-a-public (rtb-0ae899106d6e39a79) is explicitly associated with subnet-010a5338954105802 / ewb-a-public. Subnet IDs for the two private tables are truncated in the console view.

Checkpoint 02 – EC2 Instances

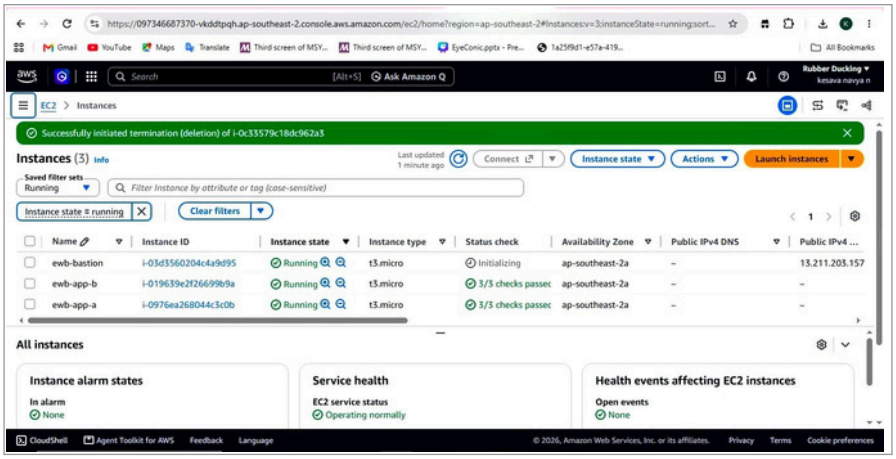


Figure 2. Checkpoint 02 – EC2 Instances (3 running, t3.micro, ap-southeast-2a): ewb-bastion (i-03d3560204c4a9d95, status check Initializing, public IPv4 13.211.203.157), ewb-app-b (i-019639e2f26699b9a, 3/3 checks passed) and ewb-app-a (i-0976ea268044c3c0b, 3/3 checks passed). The banner shows termination was initiated for a further instance, i-0c33579c18dc962a3, which is not part of the running set.

Note: the Private IPv4 column is not visible in this screenshot. Private addresses used in this report (App-A 10.10.2.10, App-B 10.20.1.10) are read from the terminal prompts in Figures 7–9. The bastion's private address is not shown directly; the App-A login record in Figure 9 lists 10.10.1.10 as the previous login source, consistent with the bastion. No Elastic IP is shown **[PARTIAL EVIDENCE: private-IP column]**.

Checkpoint 03 – Active VPC Peering Connection

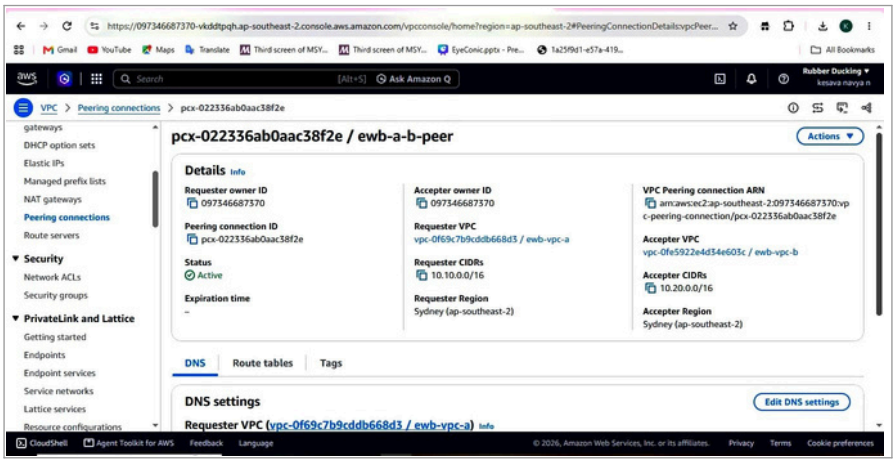


Figure 3. Checkpoint 03 – Peering connection pcx-022336ab0aac38f2e / ewb-a-b-peer, status **Active**. Requester VPC vpc-0f69c7b9cddb668d3 (ewb-vpc-a, 10.10.0.0/16); accepter VPC vpc-0fe5922e4d34e603c (ewb-vpc-b, 10.20.0.0/16); both in Sydney (ap-southeast-2); requester and accepter owner ID 097346687370.

Checkpoint 04 – Updated Route Tables with Cross-VPC Routes

Routetable	ID	Destination	Target	Status
ewb-rt-a-private	rtb-0336010ae588b4b80	10.10.0.0/16 | 10.20.0.0/16	local | pcx-022336ab0aac38f2e	Active | Active
ewb-rt-a-public	rtb-0ae899106d6e39a79	0.0.0.0/0 | 10.10.0.0/16 | 10.20.0.0/16	igw-0048ae885e7134e09 | local | pcx-022336ab0aac38f2e	Active (all)
ewb-rt-b-private	rtb-0099e9dbf5eb690b3	10.10.0.0/16 | 10.20.0.0/16	pcx-022336ab0aac38f2e | local	Active | Active

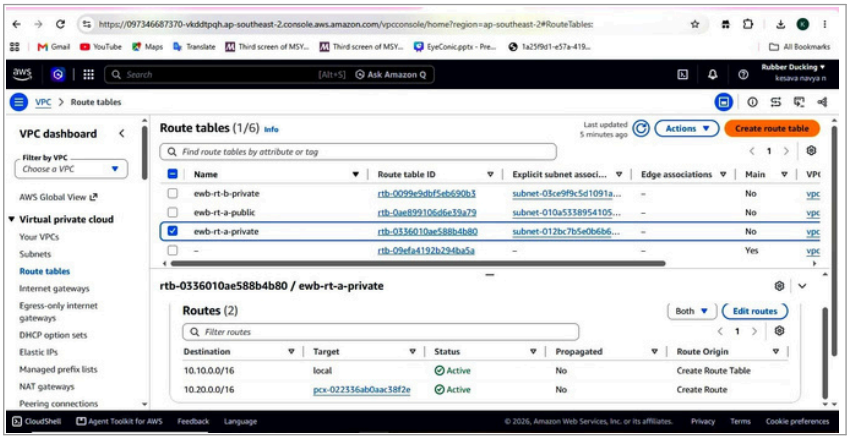


Figure 4. Checkpoint 04 (1/3) – ewb-rt-a-private: 10.10.0.0/16 local; 10.20.0.0/16 to pcx-022336ab0aac38f2e (Active).

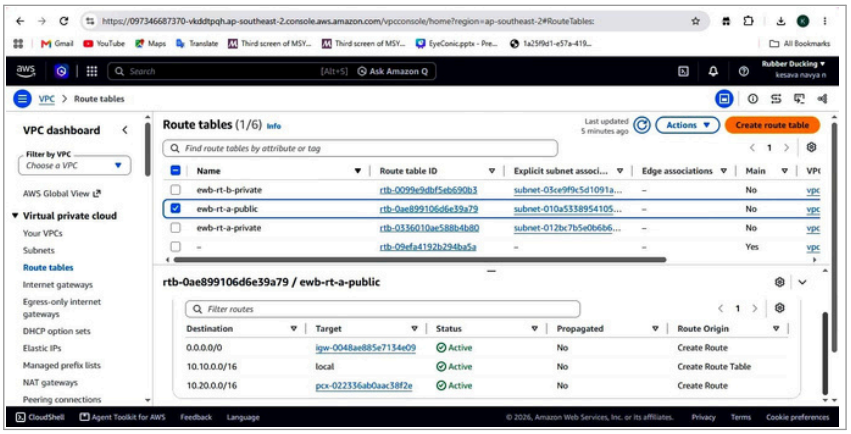


Figure 5. Checkpoint 04 (2/3) – ewb-rt-a-public: 0.0.0.0/0 to igw-0048ae885e7134e09; 10.10.0.0/16 local; 10.20.0.0/16 to pcx-022336ab0aac38f2e (all Active).

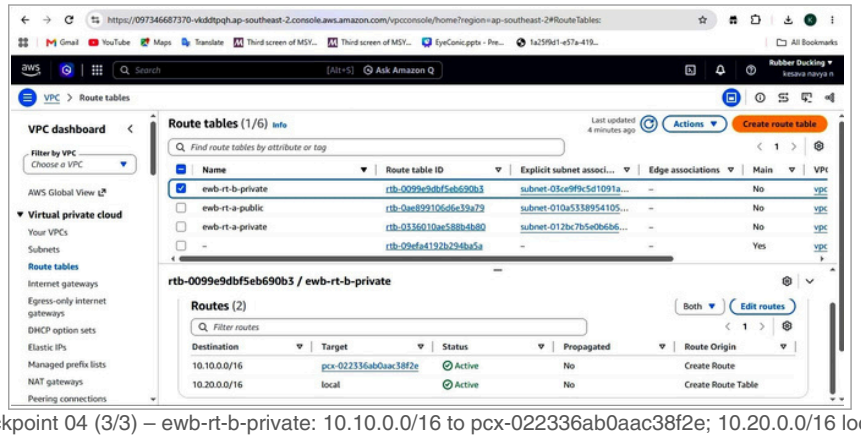


Figure 6. Checkpoint 04 (3/3) – ewb-rt-b-private: 10.10.0.0/16 to pcx-022336ab0aac38f2e; 10.20.0.0/16 local (all Active).

5. Connectivity Tests (Tests 01–05)

Test	Description	Sourceto destination	Result (from evidence)
01	App-A to App-B ICMP	10.10.2.10 to 10.20.1.10	4/4 received, 0% loss; RTT min/avg/max/mdev 0.292/0.324/0.401/0.044 ms
02	App-B to App-A ICMP	10.20.1.10 to 10.10.2.10	4/4 received, 0% loss; RTT 0.263/0.282/0.315/0.019 ms
03	SSH via ProxyJump to App-B	Client to bastion to 10.20.1.10	Host-key prompt via proxy command, login to Amazon Linux 2023 shell
04	Controlled ICMP failure	10.10.2.10 to 10.20.1.10	4 transmitted, 0 received, **100% loss**
05	ICMP recovery		

Test 01 – App-A to App-B | Test 02 – App-B to App-A | Test 03 – SSH ProxyJump to App-B

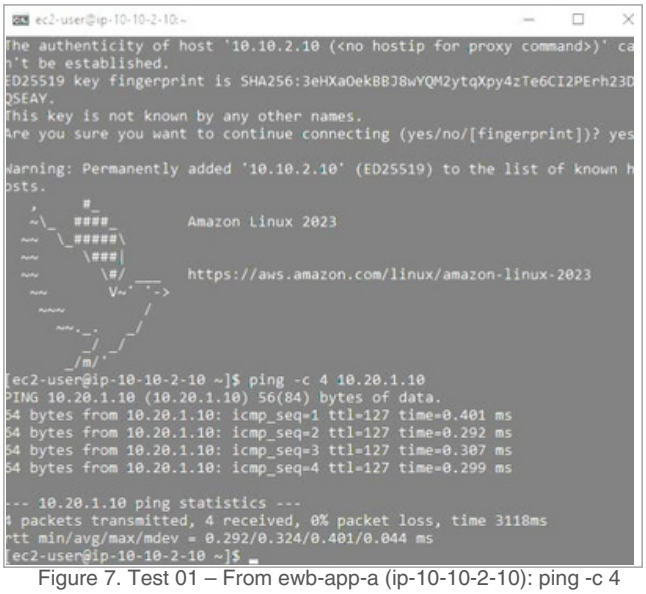


Figure 7. Test 01 – From ewb-app-a (ip-10-10-2-10): ping -c 4 10.20.1.10; 4 packets transmitted, 4 received, 0% packet loss, time 3118 ms.

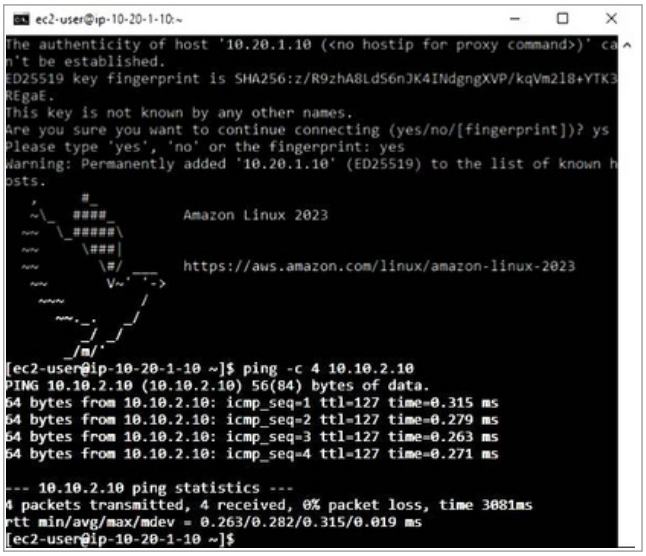


Figure 8. Tests 02 and 03 – Top: SSH to 10.20.1.10 through the proxy command (host-key prompt, Amazon Linux 2023 login). Bottom, from ewb-app-b (ip-10-20-1-10): ping -c 4 10.10.2.10; 4 received, 0% packet loss, time 3081 ms.

Test 04 – Controlled ICMP Failure | Test 05 – ICMP Recovery

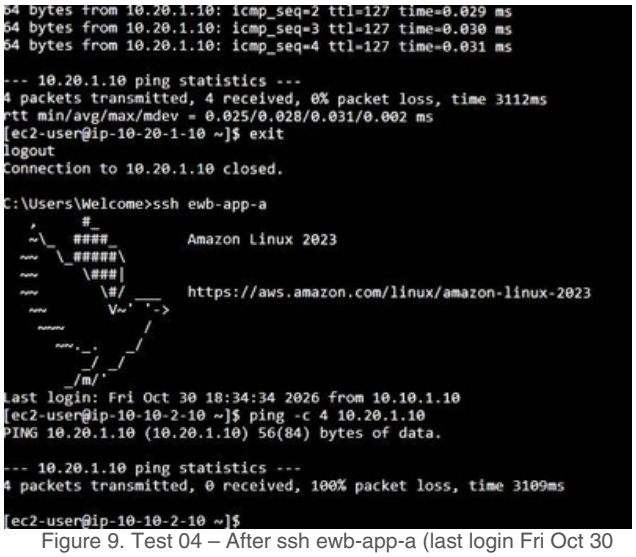
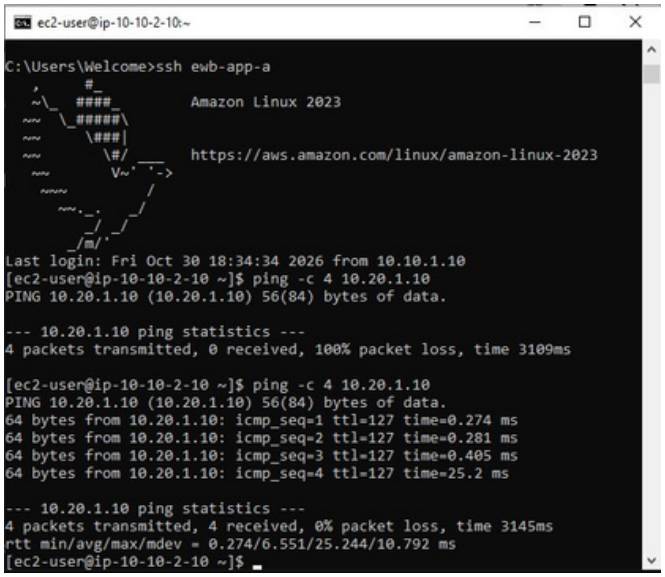


Figure 9. Test 04 – After ssh ewb-app-a (last login Fri Oct 30 18:34:34 2026 from 10.10.1.10), ping -c 4 10.20.1.10 from ip-10-10-2-10: 4 transmitted, **0 received, 100% packet loss**, time 3109 ms. Lines above show the tail of an earlier ping and the logout from the App-B session.



The supplied screenshots do not show how the failure in Test 04 was induced or reversed (for example a security-group or route change), so no cause is asserted **[MISSING EVIDENCE: fault-injection and recovery steps]**.

6. Evidence Validation Summary

#	Evidence item	Figure	Status
1	Checkpoint 01 – Route tables, explicit subnet associations	Fig. 1	**Present**
2	Checkpoint 02 – EC2 instances (names, status, IPs)	Fig. 2	**Present** **Present**
3	Checkpoint 03 – Active peering ewb-a-b-peer	Fig. 3	**Present**
4	Checkpoint 04 – Route tables with pcx routes	Figs. 4–6	**Present**
5	Test 01 – App-A to App-B, 0% loss	Fig. 7	**Present**
6	Test 02 – App-B to App-A, 0% loss	Fig. 8	**Present**
7	Test 03 – SSH ProxyJump to App-B	Fig. 8	**Present**
8	Test 04 – ICMP failure, 100% loss	Fig. 9	**Present**
9	Test 05 – ICMP recovery, 0% loss	Fig. 10	

7. Conclusion

The evidence shows an Active peering connection (pcx-022336ab0aac38f2e) between ewb-vpc-a (10.10.0.0/16) and ewb-vpc-b (10.20.0.0/16) in ap-southeast-2, with reciprocal routes in the route tables. Bidirectional ICMP succeeded with 0% packet loss (Tests 01 and 02), private instances were reached through a bastion using SSH ProxyJump (Test 03), and a 100% loss condition was observed in Test 04, demonstrating that connectivity depends on the supporting configuration. Test 05 (recovery) is not evidenced and should be supplied before final submission; security-group rules and the Test 04 fault-injection step are likewise not captured.

8. Learning Outcomes

- Plannon-overlapping CIDR ranges (10.10.0.0/16 and 10.20.0.0/16) as a prerequisite for VPC peering.
- Create and verify a peering connection and understand that it is non-transitive and needs routes on both sides.
- Configure route tables with explicit subnet associations, and distinguish public (IGW) from private routing.
- Reach private instances securely through a bastion host using SSH ProxyJump.
- Validate connectivity with ICMP, and use a controlled failure to show how routes and security rules affect traffic.
- Document results using only verifiable evidence and flag gaps explicitly.